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SECTION A - BIOLOGY

(16 Questions = 30 Marks)

1.	Which allotrope of carbon is a good conductor of electricity? **(A)** Diamond **(B)** Graphite **(C)** Fullerene **(D)** Coal	1 M
2.	Which part of the human eye controls the amount of light entering the eye? **(A)** Cornea **(B)** Iris **(C)** Retina **(D)** Pupil	1 M
3.	Which of the following metals is a liquid at room temperature? **(A)** Iron **(B)** Mercury **(C)** Copper **(D)** Aluminium	1 M
4.	What determines the sex of a child in human beings? **(A)** The temperature at which the fertilised egg is kept **(B)** Whether the child inherits an X or a Y chromosome from the father **(C)** The mother's chromosomes alone **(D)** Random environmental factors	1 M
5.	What type of image is formed by a plane mirror? **(A)** Real and inverted **(B)** Virtual and erect **(C)** Real and erect **(D)** Virtual and inverted	1 M
6.	Which of the following organisms reproduces by budding? **(A)** Amoeba **(B)** Yeast **(C)** Plasmodium **(D)** Leishmania	1 M
7.	Which of the following is an abiotic component of an ecosystem? **(A)** Frogs **(B)** Grass **(C)** Soil **(D)** Birds	1 M

8.	Assertion (A): Burning of magnesium ribbon in air is an exothermic reaction. Reason (R): It releases heat and light during the reaction. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). **(C)** Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.	1 M
9.	Assertion (A): Incomplete combustion of a hydrocarbon results in producing soot. Reason (R): Soot is formed due to the presence of unburnt carbon particles in the flame. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). **(C)** Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.	1 M
10.	Explain how the deflection of a compass needle placed near a straight current-carrying conductor changes when the current direction is reversed. OR State Fleming's left-hand rule and describe what each finger represents.	2 M
11.	How does sexual reproduction contribute to greater variation compared to asexual reproduction?	2 M
12.	Explain the cause of hypermetropia and how it can be corrected using lenses.	2 M
13.	Why is an earth wire necessary in domestic electric circuits? Explain its role in preventing electric shocks.	3 M
14.	Why is it necessary for mammals to have oxygenated and deoxygenated blood separated in the heart?	3 M

15.	Read the following scenario: In a pond ecosystem, algae serve as producers and are consumed by small insects. The insects are eaten by small fish, and these fish are preyed upon by larger fish. Some of the waste and dead organisms are broken down by bacteria. (i) Identify the trophic levels present in this food chain. (ii) Explain how energy flows through this food chain and why the length of this food chain is limited to about four trophic levels. (iii) Describe the role of decomposers in this ecosystem. (iv) What might happen if pesticides accumulate in the aquatic plants at the base of this ecosystem? OR A student accidentally touches a hot object and quickly pulls her hand away without thinking. Describe the sequence of events occurring in the nervous system that leads to this reflex action. Explain the role of the reflex arc and identify where it is formed. Additionally, discuss why reflex actions are faster than responses involving the brain.	4 M
16.	Explain how the nature, position and size of the image formed by a concave mirror change as the position of the object varies from beyond the centre of curvature to between the pole and the principal focus. Support your answer with a labelled ray diagram for each case. OR Explain the factors affecting the resistance of a conductor and derive the mathematical expression relating resistance with length and area of cross-section of the conductor. Also define resistivity and state its significance.	5 M

SECTION B - CHEMISTRY
(13 Questions = 25 Marks)

17.	Which two wires in domestic electric circuits are primarily responsible for carrying the potential difference of 220 V? **(A)** Live wire (red) and Earth wire (green) **(B)** Neutral wire (black) and Earth wire (green) **(C)** Live wire (red) and Neutral wire (black) **(D)** Earth wire (green) and Protective wire (yellow)	1 M
18.	What is the gap between two neurons called? **(A)** Axon **(B)** Dendrite **(C)** Synapse **(D)** Impulse	1 M

19.	A person with myopia has a far point at 1.2 m. Which lens would correct this defect? **(A)** Convex lens **(B)** Concave lens **(C)** Cylindrical lens **(D)** No lens required	1 M
20.	Which of the following is a characteristic of a combination reaction? **(A)** One reactant breaks down into simpler products **(B)** Two or more reactants form a single product **(C)** Two reactants exchange ions to form two new products **(D)** One element displaces another from its compound	1 M
21.	Which of the following compounds is an unsaturated hydrocarbon? **(A)** Ethane (C2H6) **(B)** Methane (CH4) **(C)** Ethene (C2H4) **(D)** Propane (C3H8)	1 M
22.	Where does the breakdown of pyruvate to carbon dioxide and water occur in human cells during aerobic respiration? **(A)** Mitochondria **(B)** Cytoplasm **(C)** Chloroplast **(D)** Nucleus	1 M
23.	If a current is flowing in an east to west direction through a wire, what is the direction of magnetic field at a point directly below the wire? **(A)** From east to west **(B)** From west to east **(C)** Clockwise when viewed from east end **(D)** Anti-clockwise when viewed from east end	1 M
24.	Assertion (A): The filament of an electric bulb is made of tungsten. Reason (R): Tungsten has a very high melting point and does not melt easily even at high temperatures. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). **(C)** Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.	1 M
25.	The radius of curvature of a spherical mirror is 20 cm. Calculate its focal length.	2 M

26.	In the reaction 2Cu + O2 → 2CuO, identify the substance oxidised and the substance reduced. OR Explain the role of adrenaline during a 'fight or flight' response in animals.	2 M
27.	Explain the relationship between the concentration of hydrogen ions in a solution and its pH value. How does this affect the strength of acids and bases?	2 M
28.	A teenage girl notices changes such as the beginning of menstruation, breast enlargement, and hair growth in new areas of her body. Meanwhile, a teenage boy begins developing facial hair, voice changes, and experiences spontaneous erections. (i) Explain why these changes occur during adolescence. (ii) What is the term used to describe this period of sexual maturation? (iii) How do these physical changes relate to the reproductive capability of humans?	4 M
29.	Explain the sequence of events that occur during a reflex action when you accidentally touch a hot object. Include the roles of receptors, neurons, spinal cord, and muscles in your answer. OR Explain the causes, symptoms, and correction of myopia, hypermetropia, and presbyopia. Include the type of lenses used for corrections and how these lenses help in restoring normal vision.	5 M

SECTION C - PHYSICS

(10 Questions = 25 Marks)

30.	Which of the following is a characteristic of alkalis? **(A)** They produce H+(aq) ions in water **(B)** They conduct electricity in solid state **(C)** They produce OH–(aq) ions in water **(D)** They turn blue litmus paper red	1 M
31.	Which of the following represents a valid food chain? **(A)** Grass → Goat → Human **(B)** Grass → Wheat → Mango **(C)** Goat → Cow → Elephant **(D)** Grass → Fish → Goat	1 M

32.	Assertion (A): Translocation in plants occurs in the phloem tissue. Reason (R): Phloem transports products of photosynthesis from leaves to other parts of the plant. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). **(C)** Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.	1 M
33.	On what factors does the resistance of a conductor depend? List them. OR What are the major parts of the human brain and their functions?	2 M
34.	What is the role of decomposers in an ecosystem?	2 M
35.	In Mendel's experiment with pea plants having two different traits (seed shape and seed colour), how was it demonstrated that these traits are inherited independently?	3 M
36.	Explain why metals like sodium and potassium are stored under kerosene oil.	3 M
37.	Describe what happens to the uterus lining when the egg released during the menstrual cycle is not fertilised and explain why menstruation occurs.	3 M

38.	A person is suffering from acidity after overeating and is advised to take a mild base such as baking soda or milk of magnesia as a remedy. (i) Explain the chemical principle on which this remedy works. (ii) What is the role of antacids in the stomach? (iii) If the pH of the stomach is normally 2, what would be the expected pH after taking the antacid? (iv) How does this relate to the neutralisation reaction discussed in acids and bases? OR A team of students carried out a breeding experiment using pea plants to observe tallness (T) and shortness (t). They noticed that when they crossed a tall plant with a short plant, all progeny were tall. However, when these F1 tall plants self-pollinated to produce F2 generation, 75% of the plants were tall, and 25% were short. (i) What do these results indicate about dominant and recessive traits? (ii) If the tallness trait depends on a plant hormone controlled by a gene producing an enzyme, explain how different forms of the gene can lead to tall or short plants. (iii) Predict the genotypic ratio of the F2 plants based on Mendel's experiments. (iv) How does this experiment support the rule that traits are inherited independently during sexual reproduction?	4 M
39.	Explain the detailed process involved in the extraction of metals that are high up in the reactivity series. Illustrate your answer with the extraction of aluminium from its ore. OR Explain the process of neutralisation reaction between an acid and a base with an example. Describe two everyday life situations where neutralisation reactions are applied and explain their importance.	5 M